

Tamás Bukits

Computer Vision Engineer & 3D Reconstruction Researcher

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PROFILE

Industrial PhD Researcher at Vicomtech working at the intersection of computer vision, neural rendering, and extended reality, designing AI-driven pipelines that transform real-world people, objects, and environments into immersive digital experiences. My research integrates multimodal sensing, semantic scene understanding, volumetric capture, 3D Gaussian Splatting, and digital twins, with a focus on developing scalable, human-centred solutions for immersive applications.

PROFESSIONAL EXPERIENCE

Researcher, Interactive Media Feb. 2024 – Present

Vicomtech San Sebastián, Spain

- Progressed from Research Intern to Research Assistant and Industrial PhD Researcher in the Digital Media Department.
- Develop AI-driven pipelines for reconstructing real-world scenes, people, and objects into immersive XR environments.
- Research 3D reconstruction, 3D Gaussian Splatting and volumetric capture, including methods that reduce dependence on Structure-from-Motion (SfM) and COLMAP-based initialization.
- Integrate semantic segmentation and multimodal scene understanding into live and offline volumetric media workflows.
- Contribute to European and industrial R&D projects in immersive media, extended reality and human-centred XR.
- Publish and present research at international conferences as both first author and co-author.

Research 3D reconstruction, 3D Gaussian Splatting, neural rendering, volumetric capture, semantic segmentation, computer vision, computer graphics, deep learning, digital twins, extended reality

Programming Python, PyTorch, TensorFlow/Keras, C/C++, CUDA, OpenGL

Software Engineer Jan. 2022 – Jan. 2024

SwiconGroup Zrt. Budapest, Hungary

- Built backend and frontend functionality for contract-management and HR workflow products.
- Developed features for a Java-based sales management application

Programming TypeScript, Node.js, Angular, MongoDB, Java

EDUCATION

Industrial PhD in Communication Technologies and Systems Oct. 2025 – Present

Polytechnic University of Madrid Madrid, Spain

Thesis: *Bringing Real World Entities Into Immersive Environments*: Designing AI-driven pipelines that digitise real-world people, objects, and environments using multimodal sensing, 3D scene understanding, and neural rendering to create scalable, high-fidelity immersive experiences for extended reality.

Erasmus Mundus Joint MSc in Image Processing and Computer Vision Sep. 2022 – Jun. 2024

TRIPLE DEGREE

Pázmány Péter Catholic University · Autonomous University of Madrid · University of Bordeaux Hungary · Spain · France

Grade: 4.11/5.00

Thesis: *AI Enhancement of Volumetric Captures* — a Gaussian Splatting pipeline designed to reduce reliance on Structure-from-Motion and COLMAP.

BSc in Computer Science Engineering Sep. 2017 – Jun. 2021

Budapest University of Technology and Economics Budapest, Hungary

Grade: 4.49/5.00

Thesis: *Algorithms for Smoothing Polyhedra* — a framework for generating smoothly connected surfaces from control polyhedra.

RESEARCH PROJECTS

BAZKARIA

Immersive shared dining

XR telepresence research for remote dining and volumetric interaction with Mugaritz.

Role: Built a real-time point-cloud pipeline for 3D capture, transmission and display.

SCENARIST

Synthetic data for safer spaces

Photorealistic labelled datasets generated through 3D digitisation and Unreal Engine simulation.

Role: Created Gaussian Splatting assets from scans of real people and objects.

AMPLIFY

AI-generated 3D memories

Volumetric reconstruction for immersive cultural experiences at Gran Teatre del Liceu.

Role: Developed and evaluated generative-AI and Gaussian Splatting pipelines.

INSPIRE

3D assets for XR training

Digitisation of people and environments for reusable immersive-training assets.

Role: Built volumetric reconstruction workflows using Gaussian Splatting.

AUTOPILOT

Digital twins for robotic surgery

Laparoscopic 3D reconstruction supporting planning and robotic autonomy.

Role: Developed surgical reconstruction methods and digital twins.

SYNERGIA

Adaptive AI-powered XR

Personalised training and rehabilitation using generative AI and digital humans.

Role: Worked on 3D reconstruction, avatars and multi-modal XR.

PUBLICATIONS

A Sneak Peek into AI-Generated 3D Immersive Memories

ACM IMX 2026 · WiP

P. de Torres Coll, **T. Bukits**, J. Ruiz, I. Burguera Hidalgo, M. Zorrilla and S. Cabrero Barros.

The XR Table: Envisioning the Future of Remote Dining Experiences Using Immersive Telepresence

ACM DIS 2025

A. Santos-Torres, P. de Torres Coll, **T. Bukits**, R. Perisé and S. Cabrero Barros.

BAZKARIA: Eating Together Through Immersive Telepresence

EURO-XR 2025 · Poster

A. Santos-Torres, P. de Torres Coll, **T. Bukits**, A. Elosegui, I. Tamayo, R. Perisé, S. Cabrero Barros and M. Zorrilla.

Beyond Bounding Boxes: 2D Semantic Segmentation for Live Volumetric Video Streaming

ACM Multimedia 2025

T. Bukits, A. Elosegi, A. Dominguez and S. Cabrero Barros.

SPOKEN LANGUAGES

Hungarian	Native
English	Full professional proficiency
Spanish	Full professional proficiency
German	Basic

AWARDS & RECOGNITION

Jun. 2026	Selected alumni presenter at IPCVAI Day 2026	Bordeaux, France
2023 – 2024	Two Erasmus+ scholarships supporting academic exchange semesters	Madrid & Bordeaux
2022 – 2024	Image Processing and Computer Vision Excellence Scholarship - being selected in a cohort of 12 students from more than 1,000 applicants worldwide	Budapest, Hungary
2017 – 2021	Integrated MSc Scholarship - selected for a high-achieving cohort and maintained a 4.0/5.0 GPA	Budapest, Hungary